Tutorial sheet 1 – Linear Algebra

Instructions. Please prepare the following questions for your tutorial meeting in week 2. Please make sure to bring along any relevant material that may help the discussion, such as lecture notes, problem sheets etc. Solutions will be made available in week 3.

VECTOR SPACES.

1. Identify which of the structures indicated below is a vector space.

(a) $(\mathbb{R}, +, \cdot, \mathbb{R})$;

(b) $(\mathbb{C}, +, \cdot, \mathbb{C})$;

(c) $(\mathbb{C}, +, \cdot, \mathbb{R})$;

(d) $(\emptyset, +, \cdot, \mathbb{R})$;

(e) $(\mathbb{E}^3, \times, \cdot, \mathbb{R})$.

SPANNING SETS.

2. Let $V(\mathbb{F})$ be a vector space and let $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$, where $\mathbf{v}_j \in V \setminus \{\mathbf{0}\}$ for $j = 1, 2, \ldots, k$. Let $\mathbf{u} \in \text{span } S$ be a non-zero vector and define $S' := S \cup \{\mathbf{u}\}$.

True or false:

(a) span $S' =$ span S;

(b) span $S' =$ span $S +$ span $\{\mathbf{u}\}$;

(c) The set sum span $S +$ span $\{\mathbf{u}\}$ is a direct sum.

In each case, you should provide a proof, stating clearly the results you used.

3. Let U be the set of polynomials of degree n divisible by $x^2 + x + 1$.

(a) Is U a subspace of $\mathcal{P}_n(\mathbb{R})$?

(b) Find a spanning set for U when $n = 2$. Do it also for $n = 3$. Are your spans minimal?

BASES

4. Let U be the set of polynomials p of degree $n \geq 2$ satisfying $p(0) = p(1) = 0$. Show that U is a subspace of $\mathcal{P}_n(\mathbb{R})$. Find a basis for U.